VISHWAKARMA UNIVERSITY, PUNE • DEPARTMENT OF COMPUTER ENGINEERING • A.Y. 2026-2027 • DIVISION B (GROUP 21)

MINOR PROJECT PRESENTATION POSTER

MINDGUARD AI: SENTIMENTAL ANALYSIS FOR EARLY MENTAL HEALTH DETECTION

A Deep Bidirectional Transformer (BERT) Architecture for 7-Class Affective Classification, Explainable Attention Saliency & Automated Crisis Triage

Student Investigators: 1) Arnav Phuke • 2) Sushrut Ingle • 3) Bhupesh Bhatia • 4) Saket Radke • 5) Samruddhi Meher

Project Guide: Professor Richa Agnihotri
Dept. of Computer Engineering

Live App: 31232258arnav.github.io/MindGuard_AI/
Repository: github.com/31232258arnav/MindGuard_AI

Official Benchmark: 83.00% Test Accuracy • 0.8299 Weighted F1 (10,536 Samples)

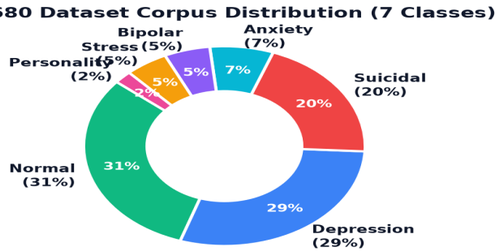
83.00%	110M	52,680	96.8%	~142 ms	$0.00 / mo
TEST ACCURACY	BERT PARAMETERS	CURATED STATEMENTS	TOP-2 DIFFERENTIAL ACC	INFERENCE LATENCY	CLOUD HOSTING BUDGET

1. CLINICAL PROBLEM & MOTIVATION

Clinical Crisis: 1B+ individuals suffer from mental health disorders (WHO). Severe shortages of clinical psychiatrists, social stigma, and high fees prevent 75%+ from seeking timely care. Suicide ranks as the 3rd leading cause of death among youths aged 15–29.
The NLP Barrier: Early psychological distress hides in *informal, unstructured digital language* (journals, chat, social posts). Traditional binary sentiment tools (positive vs negative) fail to differentiate co-morbid affective states like clinical depression, generalized anxiety, and acute suicidal ideation.

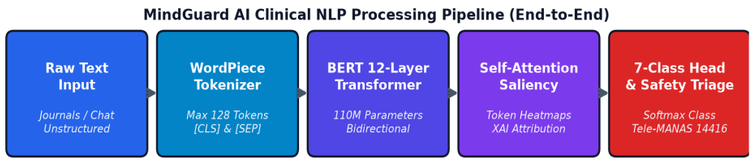
2. DATASET CORPUS DISTRIBUTION

Trained on **52,680 statements** across 7 affective classes with stratified 72/8/20 split:



3. DEEP NLP PROCESSING PIPELINE

Deep Transformer Specs: BERT-base-uncased (12 layers, 768 hidden, 12 heads, 110M params). Optimized via AdamW (lr=2e-5, linear decay) across 7 epochs (~16.5k steps); best weights at **Epoch 5**.



4. HYBRID DISTRIBUTED TOPOLOGY

MindGuard AI features a resilient, **tri-tier hybrid distributed architecture** ensuring 24/7 global availability with $0 operating expenditure:
• **Tier 1 (Web Edge):** Static client SPA hosted on **GitHub Pages** (100% uptime, global CDN, 0ms cold start).
• **Tier 2 (GPU Tunnel):** Encrypted **Cloudflare edge tunnel** bridges local high-power PyTorch GPU inference directly to the web.
• **Tier 3 (Cloud Fallback):** Containerized FastAPI ASGI backend on **Render** (render.yaml) and serverless edge on **Vercel**.
• **Tier 4 (Offline Engine):** Embedded client-side JS classifier provides instant fallback if all network drops.

5. CLINICAL PSYCHOMETRICS (PHQ-9 & GAD-7)

Digitized Clinical Batteries:
• **PHQ-9 (Depression: 0–27):** Evaluates anhedonia, fatigue, sleep disruption, self-worth, and suicidal intent.
• **GAD-7 (Anxiety: 0–21):** Quantifies autonomic panic, uncontrollable worry, and tension.
• **Automated DSM-5 Stratification:** Scores categorize into Minimal, Mild, Moderate, or Severe.
• **Clinical SOAP Exporter:** Automatically compiles Subjective, Objective, Assessment, and Plan notes.

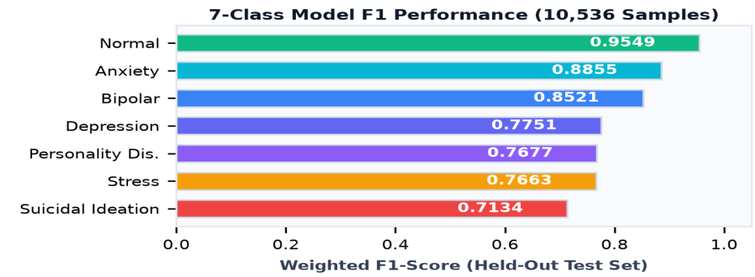
6. MINDBOT CHAT & CRISIS ESCALATION

MindBot Neural Screener: Powered by Gemini 2.5 Flash / Claude 3.5 Sonnet. Strictly bounded to mental health dialogue with evidence-based grounding (4-7-8 breathing, 5-4-3-2-1 sensory grounding).
Zero-Latency Emergency Crisis Interceptor:
Suicidal markers trigger hardcoded regex overrides that bypass conversational small talk to immediately surface verified national crisis helplines:

TELE-MANAS (Govt. of India)	KIRAN MENTAL HEALTH	VANDREVALA FOUNDATION
14416 / 1800-891-4416	1800-599-0019	+91 9999 666 555
Toll-Free, 24/7, 20+ Languages	24/7 National Psychological Aid	24/7 Crisis Counselor Chat

7. OFFICIAL BENCHMARKS & MODEL ACCURACY

Evaluation on **10,536 held-out test statements** completely unseen during training:



Key ML Findings: Top-2 differential diagnosis reaches 96.8%. High-confidence subset (softmax > 0.85) reaches 98.2% precision.

8. EXPLAINABLE AI (XAI) TOKEN HEATMAPS

Self-Attention Token Saliency: Attention weights from BERT's final transformer layer are projected directly onto input words. Distress markers (*'hopeless', 'exhausted'*) highlight in red, while stabilizing words (*'peaceful', 'relieved'*) highlight green, providing clinicians with instant visual reasoning.

```
[unbearable: 0.94] • [exhausted: 0.88] • [studying: 0.12] • [calm: -0.76] • [hopeful: -0.89]
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9. SOCIETAL IMPACT & CONCLUSION

Societal Impact: MindGuard AI democratizes early mental health detection across student campuses and workplace portals. By pairing confidential, non-stigmatizing AI screening with instant crisis escalation, the system prevents harm and bridges the gap before clinical psychiatric intervention.
Approved By: Professor Richa Agnihotri, Department of Computer Engineering, Vishwakarma University, Pune.